

Neuromorphic Human-Computer Interaction: A Wikipedia Study

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CCS CONCEPTS

• **Computer systems organization;**

KEYWORDS

Neuromorphic Computing, Some more keywords

1 INTRODUCTION

Browsing the internet has become an integral part of everyday modern life. The internet is often subdivided into connected information networks that humans navigate using their internal models of the world. Understanding how humans navigate through such information networks gives insight into the connectedness of topics from the viewpoint of users. With this knowledge of semantic relations, the curators of these networks can organise their network nodes in a way that allows for more efficient and intuitive user navigation.

Wikipedia is a well-known example of a virtual information network that can be represented as a directed graph with articles as nodes and links between articles as the edges. It is partially structured by the semantic relatedness of the articles, and therefore acts as an approximation of humans' semantic world models. However, links between articles are added at the discretion of the article's author. While these links are often strongly semantically related, the amount of links and the degree of relatedness between linked articles can vary wildly. This makes it difficult to derive semantic relations from graph structure alone.

Wikispeedia is a game where the goal is for users to navigate from a source to a target article in a small version of Wikipedia (cite stuff here). Data collected from this game offers insight into the semantic relatedness of articles without direct connections, as well as the degree of degree of relatedness (the semantic distance) between articles. West et al. (cite wikispeedia article) demonstrated that semantic distance between concepts can be inferred from this game data.

In the interest of efficient navigation, it is worthwhile to investigate the relationship between the semantic distances of concepts and navigation speed. Understanding how semantic distance affects the time taken to move from a source to a target article offers opportunities to improve graph organisation. For example, rather than only having links added by authors, Wikipedia could provide additional links to concepts that minimises navigation time. This intuition can also be applied to the organisation of other semantic information networks.

We aim to investigate this relationship by analysing Wikispeedia game data. However, due to the graph structure of Wikipedia, semantic distance is not the only factor influencing the duration of games. Factors such as the in-degree of an article (the amount of directed edges that have that article as the end node) also impact navigation difficulty. Defining an index of difficulty for Wikispeedia games that accounts for semantic distance as well as other confounding factors can be an initial step in investigating this relationship.

2 BACKGROUND

In this section we provide an overview of background literature in current academic field.. Through this, we aim to embed our project in this field, and present the foundations of our research.

2.1 Human navigation in information space

In this paper we investigate navigation in information space, specifically targeted navigation from some starting-page to a goal-page. Specifically we are interested in investigating whether movement in this information space is similar to movement in physical space.

2.1.1 Signal-to-noise ratio. Shannon's Theorem describes the information capacity C (in bits/s) of a communications channel of bandwidth B as

$$C = B \log_2(SN + 1)$$

. Fitts proposed an index of difficulty (ID) for a target acquisition task where signal power (S) and noise power (N) are replaced by movement amplitude (A) and target width (W), respectively. This was later reformulated to the Shannon notation, as:

$$ID = \log_2\left(\frac{A}{W} + 1\right)$$

According to MacKenzie "In Fitts' analogy, movement amplitudes are like signals and target tolerances or widths are like noise.[6]". In information space, a similar analogy seems sensible. Here, movement amplitudes could be computed not over physical distances, but over some distance over the two objects in the information space. The noise of an object would be the target tolerance, which in networks would be the ease of access of a node.

2.2 Semantic knowledge from networks

Proposed computing the semantic distance as the path length between two concepts in a concept graph [4]. [TODO] The analysis of Wikipedia's structure as a semantic knowledge graph is quite well established. Several works investigate the possibility of extracting

additional information on the semantic distance between concepts from Wikipedia, all of these show promising results[2, 3, 5]. These findings suggest a relationship between the semantic similarity of pages, and their distance in the Wikipedia graph-structure. [extend?]

2.3 Similarity from Wikipedia traces

The papers mentioned above mostly focus on information that can directly be extracted from the structure of data within Wikipedia. However, there are other aspects to consider as well, such as the way this Wikipedia data-structure is actually utilized by its users. West et al. [14] investigate the possibility of extracting semantic distances from a dataset of Wikispeedia games. This dataset consists of network traversal paths, where each path represents a person playing the Wikipedia game. In this game, a user is tasked to navigate from source-page A to goal-page B in the least amount of steps, thus these traces show the behaviour of targeted navigation on the Wikipedia network. The most relevant finding in this paper is that these traces can be analysed to determine semantic distances between concepts that are sensible when presented to human evaluators, improving over an LSA baseline. This approach was extended in 2013 by Singer et al.[11] to include distances for items which don't directly occur in the same path, and used a larger dataset. A 2016 paper [7] furthermore finds that even traces from unconstrained navigation can be used to infer semantic similarity between topics.

2.4 Navigating Wikipedia using semantic information

In a 2016 paper, Nogueira and Cho apply these concepts to create a model which produces random walks between nodes on the Wikipedia graph[8]. They reference the work by West et al. stating that they “see that the goal-driven web navigation is a generalization and re-framing of the Wikispeedia”. Their approach places this concept of semantic navigation at its core, representing semantic similarity as a small distance between bag-of-words and attention mechanism embeddings. The model uses beamsearch on these embeddings to determine it's next move through a Long Short-Term Memory network. They find that the model outperforms several baselines on the WikiNav-Jeopardy benchmark.

In 2022 Zaheer et al. extend the approach outlined by Nogueira and Cho by using transformer models for embedding rather than a bag-of-words approach, and removing complex search at evaluation time[15]. These changes allowed them to use a significantly larger graph. This paper massively increased the likelihood of successfully navigating between points A and B, from 12.5% to 90%. Zaheer et al. demonstrate the applicability of this model in performing targeted navigation on information graphs using this semantic exploration approach by applying it on a fact-checking and question answering benchmark. Their method vastly improves baselines, at times approaching or improving over state-of-the-art methods. The paper by Zaheer et al. shows strong evidence that semantic similarity extracted by transformers facilitate a good navigation policy “Using random features in place of [transformers] caused the performance to drop to near chance, showing that the models

are utilizing the semantics at each node and are not relying on a general search strategy of some kind”.

2.5 How humans navigate Wikipedia

Another interesting result from the 2009 paper by West et al. [14] is the fact that targeted human wayfinding in Wikipedia consists of two phases: a generalization phase and a specification phase. Since source-page A and goal-page B don't necessarily have a clear human-sensible path between the two directly, players of the game often first navigate to a “hub”-page, which have a large number of outgoing edges to different topics. West et al. investigate such strategies further in [13], describing several strategies and developing a target prediction model based on these. A more thorough analysis of user behaviour on Wikipedia is conducted by Piccardi et al. in 2023, and again by Piccardi and West in 2025 [9, 10]. They find that Wikipedia usage does not always follow the same structure as targeted search, meaning results do not generalize to all Wikipedia usage. Specifically natural Wikipedia usage is often very short, only one (or very few) clicks. This can be explained by viewing Wikipedia in the context of the entire Web. Exploration often happens outside of Wikipedia, with users querying search engines such as Google for information, Google then in turn returns Wikipedia pages containing relevant information.

Might be interesting:

3 METHODS

In order to try and find useful difficulty metrics for Wikispeedia, we compared various potential metrics to see how well they predict the ‘real’ difficulty, measured by how long the user ended up taking. To do this, we tried the same formula as Fitts’ law, $ID = \log_2(\frac{D}{W} + 1)$, by investigating metrics that are conceptually similar to distance (how far we are removed from our target page) and width (how easy it is to reach the target page).

Furthermore, we used binning to group tasks with similar difficulty. The reason for this is that without binning, we have a lot of tasks that are similar in difficulty but have very few samples, causing a lot of noise. We primarily used proportional bins (where we vary bin width and attempt to balance the number of data points per bin) in this paper, but the appendix also contains results without binning (where each datapoint represents a single task with at least three datapoints) and results with fixed-width bins.

After binning, we use linear regression to see if there is a correlation between the index of difficulty and the number of clicks taken. For this, we used SciPy [12].

3.1 Distance metrics

In this paper, we tried the following three distance metrics: *semantic distance*, *semantic graph distance*, and *graph distance*.

The first distance metric, *semantic distance*, is calculated by the LLM transformer model [?] and indicates the similarity of the titles of two Wikipedia articles.

In order to calculate the second distance metric, *semantic graph distance*, we first construct a graph of all Wikipedia pages in the dataset. The articles represent the vertices in the graph, and the links between them are the edges. We then use the LLM’s semantic

distance as the length of each edge. The semantic graph distance between two articles is the length of the shortest path in this weighted graph.

The last distance metric, *graph distance*, represents the length of the shortest path between two pages, but where all edges simply have a weight of 1, representing the minimal number of clicks.

3.2 Width metrics

The width-like metric we used is *PageRank*. To explain the intuition behind PageRank, assume a user who is surfing Wikipedia. This user starts on some random page and randomly clicks a link. At some point this user gets bored and starts at some other random page and continues clicking links randomly. The PageRank of page a is defined as the chance this user reaches it [1]. Therefore, the higher the PageRank, the easier it is to reach the article, similar to how a wider button is easier to reach, as PageRank grows with more incoming links, and gives more weight to incoming links from pages that are themselves easy to reach.

We also tried using the indegree of a page in the Wikipedia graph, but found that it gave strictly worse results. We will not discuss indegree as a metric further.

3.3 Binning strategies

The dataset we use is quite noisy and contains many entries with slightly different distances and widths. By binning the data on the ID we reduce noise and make the relation more smooth. We tried binning the data in two different ways: *fixed width binning* and *proportional binning*.

For fixed width binning, all bins have an equal ‘width’. Since all points are not distributed equally over the ID , bins contain big differences in the number of points they contain.

For proportional binning, bins contain the same number of points. Since all points are not distributed equally over ID , the ranges of the bins also differ significantly.

4 RESULTS

We present our results in Figure 1 and Figure 2. These figures differ in the binning strategy used: Figure 1 uses proportional binning and Figure 2 uses fixed size binning. We will go over the various results we have, grouping them by the used distance metric.

Semantic distance. We found that the semantic distance by itself is a relatively poor indicator of task difficulty, with $r = 0.56$ and $r = 0.70$. However, by including the PageRank of the target article, results improved significantly, going up to $r = 0.84$ and even $r = 0.89$. We believe the reason for this is that the semantic distance by itself does not take into account the graph structure, which seems to be an important factor in total task difficulty. Adding PageRank ensures both of these are factored in.

Semantic graph distance. The results here are good, giving $r = 0.91$ for proportional binning and $r = 0.79$ for fixed-size binning without PageRank. This further reaffirms our earlier belief that graph structure is important: this distance metric includes the graph structure by definition. Adding PageRank gives us $r = 0.87$ for both binning methods, which is a relatively small difference, giving a small improvement for one binning type but a small regression for

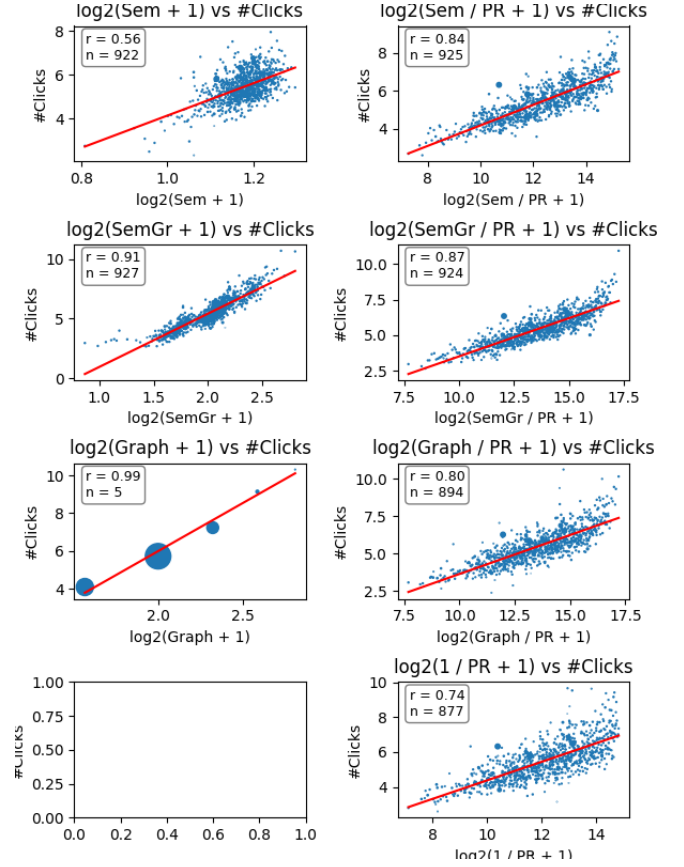


Figure 1: Number of used clicks plotted against difficulty, using various difficulty measures and proportional binning. Left: without PageRank. Right: with PageRank.

Top: ‘direct’ semantic distance. Top center: ‘graph’ semantic distance. Bottom center: graph distance. Bottom: without distance metric.

the other. Overall it seems that adding PageRank here thus does not contribute much.

Graph distance. These results are very limited when we do not include PageRank, as they collapse into 5 or 6 bins. This is because there are only very few possible values here. As such the r -values of $r = 0.99$ and $r = 0.97$ look nice, but are kind of ‘inflated’ by the low number of datapoints. However, it is interesting to see that the relationship does seem to hold according to the limited information we have.

When we add PageRank, which gives us a reasonable number of bins again, we find $r = 0.80$ and $r = 0.81$. This is lower than the other two distance metrics, which seems to indicate that the semantic distance adds useful information about the difficulty of a task.

No distance. Finally, using only PageRank as a difficulty metric gives mediocre results, with $r = 0.74$ and $r = 0.73$. Given that this metric only looks at the target article and not the source article, we think it would be unreasonable to expect better; in fact it is worse

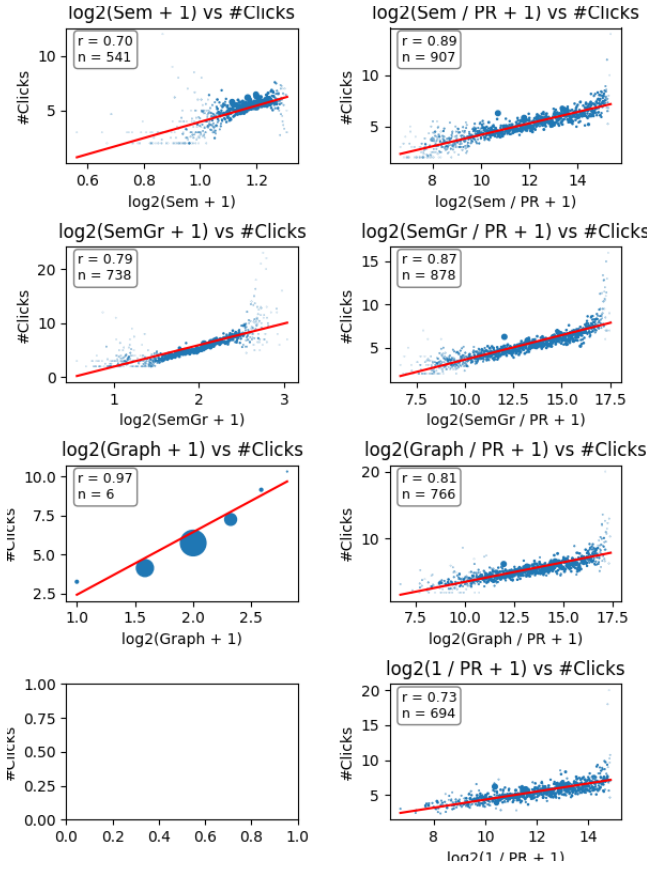


Figure 2: Number of used clicks plotted against difficulty, using various difficulty measures and fixed-width binning. Left: without PageRank. Right: with PageRank. Top: ‘direct’ semantic distance. Top center: ‘graph’ semantic distance. Bottom center: graph distance. Bottom: without distance metric.

than all other metrics we tried, other than semantic distance which ignores the graph structure completely.

Of course, there are no results for not using a distance metric and also ignoring PageRank, as we then do not have a difficulty metric at all. This explains the empty graphs in our figures.

5 DISCUSSION

Metric comparison. Overall, our results seem to indicate that multiple factors are important for the difficulty of a Wikispeedia task. The best results are achieved by metrics that take into account a combination of semantic distances and graph structure, while metrics that completely ignore one in favour of the other give worse results.

There is no one best difficulty metric in the results we found, but there are multiple that look promising and seem to give a reasonable indication of the number of clicks a user will require.

Tail end. It also looks like the ‘tail end’ (high difficulty) tasks have a slightly stronger-than-linear upwards trend. However, the

data is too sparse and noisy to draw a clear conclusion about this: it can mean that there is no ‘true’ log-linear relationship, but it can also mean that these high-difficulty tasks ‘overwhelm’ the user, and may be similar to discontinuities seen in Fitts’ law. (TODO: leave this in? and add citation if yes.)

Binning methods. The results for the two binning methods are roughly similar, but differ quite a bit in details. For example, the combination of semantic graph distance and PageRank is the best metric when evaluated using proportional binning, but relatively poor when evaluated using fixed-width binning. Looking at the scatterplots, we hypothesize that this may be related to outliers in these metrics. The graph distance, for example, only has a small number of datapoints with 5 or 6 clicks, and this likely extends to the semantic graph distance having outliers as well. Proportional binning groups these together, thus reducing their impact. On the other hand, semantic distance and PageRank are less likely to have such outliers, which likely explains the good results of this combination when using fixed-width binning.

REFERENCES

- [1] Sergey Brin and Lawrence Page. 1998. The Anatomy of a Large-Scale Hypertextual Web Search Engine. *Computer Networks and ISDN Systems* 30, 1–7 (1998), 107–117. [https://doi.org/10.1016/S0169-7552\(98\)00110-X](https://doi.org/10.1016/S0169-7552(98)00110-X)
- [2] Damaris Fuentes-Lorenzo, Jorge Morato, Sonia Sanchez-Cuadrado, and Luis Sanchez. 2019. Building concept maps by adapting semantic distance metrics to Wikipedia. *Education for Information* 35, 3 (2019), 209–240. <https://doi.org/10.3233/EFI-190279> arXiv:https://journals.sagepub.com/doi/pdf/10.3233/EFI-190279
- [3] Saket Karve, Vasisht Shende, and Swaroop Hopale. 2019. Semantic Relatedness Measurement from Wikipedia and WordNet Using Modified Normalized Google Distance. In *Data Analytics and Learning*, P. Nagabhushan, D. S. Guru, B. H. Shekar, and Y. H. Sharath Kumar (Eds.). Springer Singapore, Singapore, 143–154.
- [4] Yoed N Kenett, Efi Levi, David Anaki, and Miriam Faust. 2017. The semantic distance task: Quantifying semantic distance with semantic network path length. *Journal of Experimental Psychology: Learning, Memory, and Cognition* 43, 9 (2017), 1470.
- [5] Fei Li, Lejian Liao, Lanfang Zhang, Xinhua Zhu, Bo Zhang, and Zheng Wang. 2020. An Efficient Approach for Measuring Semantic Similarity Combining WordNet and Wikipedia. *IEEE Access* 8 (2020), 184318–184338. <https://doi.org/10.1109/ACCESS.2020.3025611>
- [6] I Scott MacKenzie. 2018. Fitts’ law. *The wiley handbook of human computer interaction* 1 (2018), 347–370.
- [7] Thomas Niebler, Daniel Schlör, Martin Becker, and Andreas Hotho. 2016. Extracting Semantics from unconstrained navigation on wikipedia. *KI-Künstliche Intelligenz* 30, 2 (2016), 163–168.
- [8] Rodrigo Nogueira and Kyunghyun Cho. 2016. Webnav: A new large-scale task for natural language based sequential decision making. *arXiv preprint arXiv:1602.02261* (2016).
- [9] Tiziano Piccardi, Martin Gerlach, Akhil Arora, and Robert West. 2023. A Large-Scale Characterization of How Readers Browse Wikipedia. *ACM Trans. Web* 17, 2, Article 11 (April 2023), 22 pages. <https://doi.org/10.1145/3580318>
- [10] Tiziano Piccardi and Robert West. 2025. Navigating Knowledge: Patterns and Insights from Wikipedia Consumption. *arXiv preprint arXiv:2501.00939* (2025).
- [11] Philipp Singer, Thomas Niebler, Markus Strohmaier, and Andreas Hotho. 2013. Computing semantic relatedness from human navigational paths: A case study on wikipedia. *International Journal on Semantic Web and Information Systems (IJSWIS)* 9, 4 (2013), 41–70.
- [12] Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, Matt Haberland, Tyler Reddy, David Cournapeau, Evgeni Burovski, Pearu Peterson, Warren Weckesser, Jonathan Bright, Stéfan J. van der Walt, Matthew Brett, Joshua Wilson, K. Jarrod Millman, Nikolay Mayorov, Andrew R. J. Nelson, Eric Jones, Robert Kern, Eric Larson, C J Carey, Ilhan Polat, Yu Feng, Eric W. Moore, Jake VanderPlas, Denis Laxalde, Josef Perktold, Robert Cimrman, Ian Henriksen, E. A. Quintero, Charles R. Harris, Anne M. Archibald, Antônio H. Ribeiro, Fabian Pedregosa, Paul van Mulbregt, and SciPy 1.0 Contributors. 2020. SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. *Nature Methods* 17 (2020), 261–272. <https://doi.org/10.1038/s41592-019-0686-2>
- [13] Robert West and Jure Leskovec. 2012. Human wayfinding in information networks. In *Proceedings of the 21st International Conference on World Wide Web (Lyon, France) (WWW ’12)*. Association for Computing Machinery, New York,

- NY, USA, 619–628. <https://doi.org/10.1145/2187836.2187920>
- [14] Robert West, Joelle Pineau, and Doina Precup. 2009. Wikispeedia: An Online Game for Inferring Semantic Distances between Concepts.. In *IJCAI*, Vol. 9. 1598–1603.
- [15] Manzil Zaheer, Kenneth Marino, Will Grathwohl, John Schultz, Wendy Shang, Sheila Babayan, Arun Ahuja, Ishita Dasgupta, Christine Kaeser-Chen, and Rob Fergus. 2022. Learning to navigate wikipedia by taking random walks. *Advances in Neural Information Processing Systems* 35 (2022), 1529–1541.

A ADDITIONAL INFORMATION

Here we give all results of the experiment. Figures 3 to 5 plot the *ID* against the number of clicks of each game, using proportional binning, fixed size binning and no binning respectively. Figures 6 to 8 plot the *ID* against the duration of each game using proportional binning, fixed size binning and no binning respectively. For each figure, we use the following partitioning. Left: without PageRank. Top: ‘direct’ semantic distance. Top center: ‘graph’ semantic distance. Bottom center: graph distance. Bottom: without distance metric.

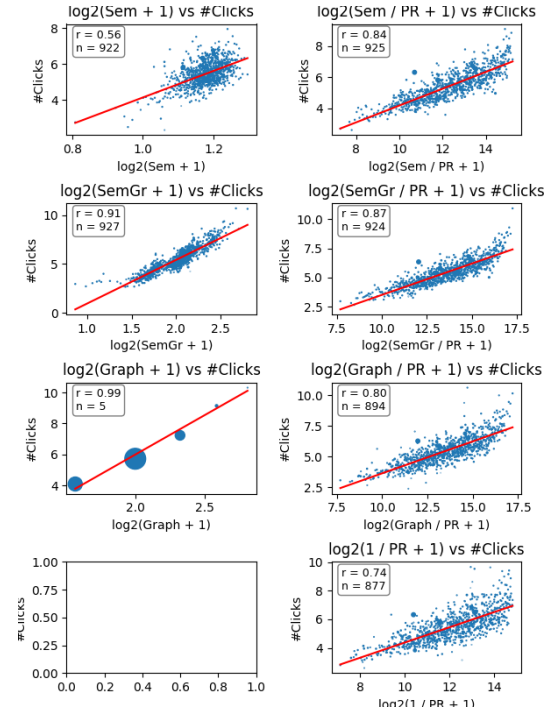


Figure 3: Number of used clicks plotted against difficulty, using various difficulty measures and proportional binning.

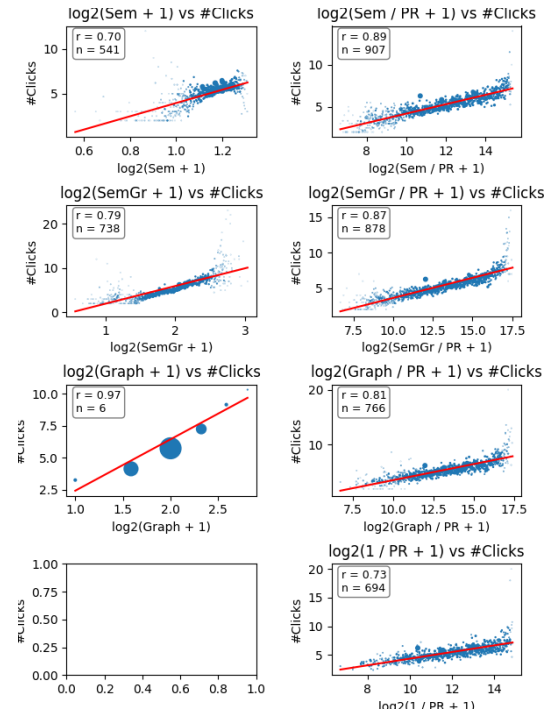


Figure 4: Number of used clicks plotted against difficulty, using various difficulty measures and fixed-width binning.

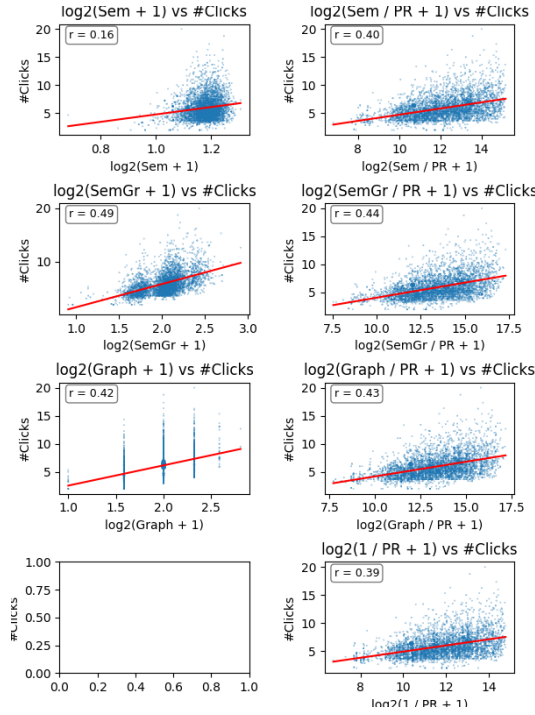


Figure 5: Number of used clicks plotted against difficulty, using various difficulty measures without any binning.

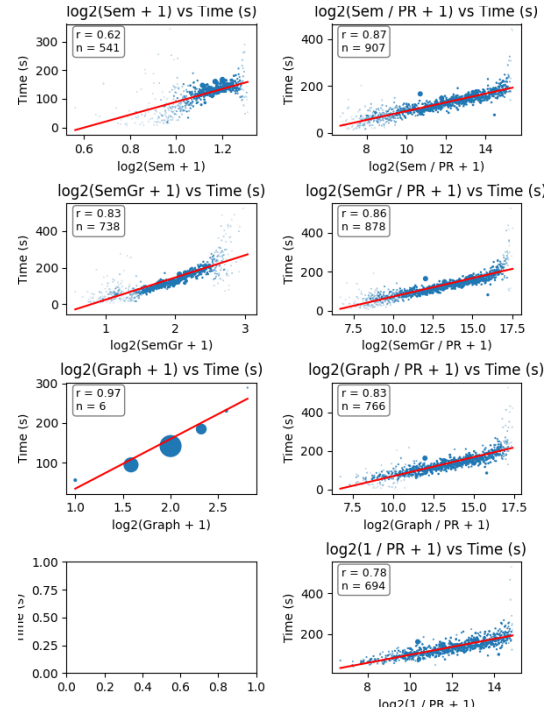


Figure 7: Duration of the game plotted against difficulty, using various difficulty measures and fixed-width binning.

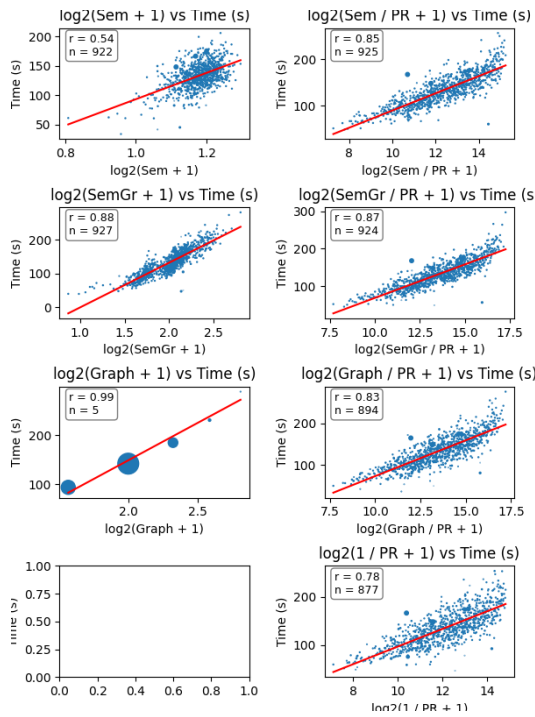


Figure 6: Duration of the game plotted against difficulty, using various difficulty measures and proportional binning.

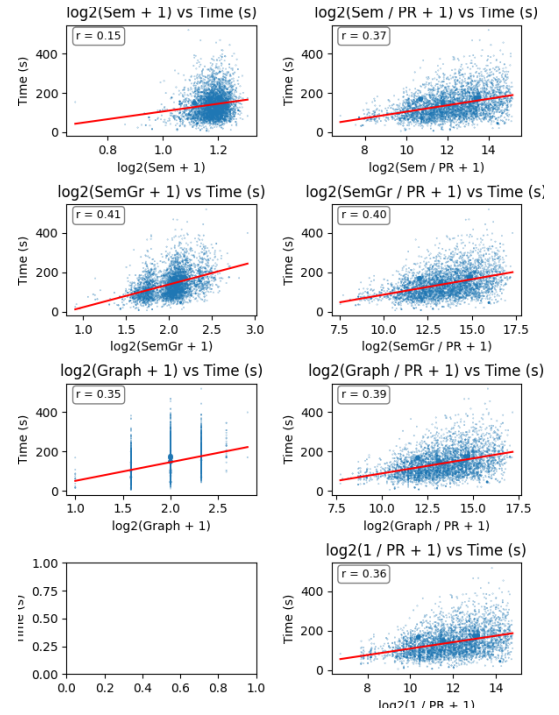


Figure 8: Duration of the game plotted against difficulty, using various difficulty measures without any binning.